E-Commerce Sales Analytics Report

Excel | Pivot Tables | Business Insights

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1. Project Overview

This project analyzes e-commerce transaction data covering 5,000 orders placed by 989 customers across four product categories and four regions, from 29 October 2013 to 29 October 2026. The goal is to uncover insights into revenue drivers, customer behavior, discount effectiveness, delivery and payment patterns, customer satisfaction, and seasonal trends, so that the business can make better strategic decisions.

The work follows an end-to-end workflow: data preparation and feature engineering on the raw order table, exploratory statistics on the numerical fields, and a set of Excel pivot-table analysis sheets (Executive Summary, Customer, Product, Time & Seasonal, and Regional analysis) that present the findings.

2. Dataset Summary

- **Rows:** 5,000 (one record per order; 5,000 unique order IDs)
- **Customers:** 989 unique customer IDs (IDs 1000 to 1999)
- **Columns:** 12 in the raw data; 20 after feature engineering
- **Order details:** order_id, order_date, product_category, region, quantity, unit_price, discount, revenue
- **Customer and service fields:** customer_id, payment_method, delivery_days, customer_rating
- **Derived fields:** YEAR, MONTHYEAR, DAYS (day of week), Day Type, SEASON, Revenue Category, DELIVERYTYPE, Rating Category
- **Missing data:** none found (0 null values, 0 duplicate order IDs)
- **Data integrity check:** revenue equals quantity × unit price × (1 − discount) on every one of the 5,000 rows
- **Tools used:** Microsoft Excel (pivot tables and slicers)

3. Exploratory Data Analysis

Data preparation

- **Data loading and structure check:** confirmed column types, a consistent date range, and no nulls.
- **Duplicate check:** every order ID is unique.
- **Revenue validation:** recalculated revenue from quantity, price and discount and matched all 5,000 rows.
- **Feature engineering:**
  - YEAR, MONTHYEAR and DAYS were extracted from the order date; Day Type labels each order Weekday or Weekend.
  - SEASON groups months into Winter, Summer, Monsoon and Autumn.
  - Revenue Category bands each order as Low (under $500), Medium ($500 to $1,500) or High (above $1,500).
  - DELIVERYTYPE bands delivery time as Express (1 to 3 days), Standard (4 to 7 days) or Slow (8 to 11 days).
  - Rating Category bands ratings as Negative (below 3), Neutral (3 to 3.9) or Positive (4 and above).

Summary statistics

Feature	Mean	Std	Min	Median	Max
quantity	4.04	2.02	1	4	7
unit_price ($)	308.42	169.26	15.15	309.89	599.96
discount	18.0%	10.1 pts	0%	18%	35%
delivery_days	6.12	3.15	1	6	11
customer_rating	2.97	1.16	1.0	3.0	5.0
revenue ($)	1,021.96	825.58	11.21	796.65	4,119.33

Table 1: Summary statistics of numerical features.

Correlations

Revenue is driven mainly by unit price (correlation 0.68) and quantity (0.62), as expected from the revenue formula. Discount has a mild negative relationship with revenue (−0.14). Customer rating is essentially uncorrelated with every other numerical field (all within ±0.02), so price, quantity, discount and delivery time do not explain how customers rate their orders.

4. Data Analysis by Business Theme

The analysis is grouped into seven themes. All figures below are calculated from the full 5,000-order table.

A. Sales and Revenue Analysis

1. Overall sales summary

Total revenue	Total quantity	Orders	Avg revenue per order	Min order	Max order
$5,109,776	20,224	5,000	$1,021.96	$11.21	$4,119.33

2. Revenue by product category

Category	Orders	Total revenue	Share	Avg order value
Electronics	1,777	$1,829,899	35.8%	$1,029.77
Clothing	1,531	$1,531,932	30.0%	$1,000.61
Home	969	$982,084	19.2%	$1,013.50
Beauty	723	$765,861	15.0%	$1,059.28

Electronics and Clothing together produce about two thirds of revenue. Beauty has the smallest volume but the highest average order value.

3. Revenue by order size band

Revenue category	Orders	Share of orders	Total revenue	Share of revenue
High (above $1,500)	1,262	25.2%	$2,797,723	54.8%

Revenue category	Orders	Share of orders	Total revenue	Share of revenue
Medium ($500 to $1,500)	1,990	39.8%	$1,859,705	36.4%
Low (under $500)	1,748	35.0%	$452,348	8.9%

A quarter of orders generate more than half of all revenue, while the 35% of orders in the Low band add under 9%.

B. Customer Behavior Analysis

4. Customer base

- 989 unique customers placed 5,000 orders, an average of 5.06 orders and $5,167 in lifetime revenue per customer.
- 96.1% of customers ordered more than once, and 55.8% ordered five or more times, so the business is driven by repeat buyers.
- The top 10% of customers (about 99 people) account for 21.1% of revenue, so revenue is fairly spread out rather than concentrated in a few accounts.

5. Top five customers by revenue

Customer ID	Orders	Total revenue
1663	14	$17,681
1955	10	$16,265
1675	11	$15,036
1276	11	$15,024
1647	11	$14,895

6. Orders per customer

Orders placed	Customers
1	39
2	91
3	149

Orders placed	Customers
4	158
5	153
6	154
7	99
8 or more	146

7. Quantity per order

Average order value rises steadily with basket size, from $258 for single-item orders to $1,769 for seven-item orders. Order counts are almost equal across quantities 1 to 7 (671 to 775 orders each), so there is no single dominant basket size.

C. Discount and Promotion Analysis

8. Discount level vs. order value

Discount band	Orders	Avg revenue per order	Avg quantity	Avg rating
0% to 10%	1,401	$1,159.46	4.01	2.96
10% to 20%	1,401	$1,057.96	4.07	2.96
20% to 30%	1,502	$943.94	4.07	2.97
30% to 35%	696	$841.07	4.00	3.04

The average discount is 18.0%. Deeper discounts lower revenue per order, but the average quantity stays flat at about 4 items, so higher discounts are not buying larger baskets. Ratings are also almost the same across bands.

D. Delivery and Payment Analysis

9. Delivery type

Delivery type	Orders	Avg delivery days	Total revenue	Avg rating
Standard (4 to 7 days)	1,843	5.52	$1,927,133	3.03

Delivery type	Orders	Avg delivery days	Total revenue	Avg rating
Slow (8 to 11 days)	1,883	9.51	$1,915,652	2.94
Express (1 to 3 days)	1,274	1.98	$1,266,991	2.95

More than a third of orders (37.7%) arrive in the Slow band, yet ratings barely move with delivery speed: average rating is 2.99 for 1-day delivery and 2.89 for 11-day delivery, with no clear trend in between.

10. Payment method

Payment method	Orders	Total revenue	Share	Avg order value	Avg rating
Card	2,270	$2,366,248	46.3%	$1,042.40	3.02
Cash on delivery (COD)	1,774	$1,788,408	35.0%	$1,008.12	2.94
Wallet	956	$955,119	18.7%	$999.08	2.93

E. Customer Satisfaction Analysis

11. Rating distribution

Rating category	Orders	Share of orders	Avg rating
Negative (below 3)	2,477	49.5%	1.96
Neutral (3 to 3.9)	1,254	25.1%	3.44
Positive (4 and above)	1,269	25.4%	4.48

The overall average rating is only 2.97 out of 5, and almost half of all orders receive a negative rating. This is the weakest area in the whole dataset.

12. Rating by category

Category	Avg rating
Clothing	3.01
Beauty	3.01
Electronics	2.95

Category	Avg rating
Home	2.94

Differences between categories, regions and payment methods are all under 0.1 points, so low satisfaction is a business-wide issue rather than a problem tied to one segment.

F. Time and Seasonal Analysis

13. Revenue by year

Year	Orders	Revenue
2013 (from 29 Oct)	68	$73,604
2014	384	$389,882
2015	384	$377,761
2016	386	$368,256
2017	384	$383,727
2018	384	$419,400
2019	385	$397,995
2020	385	$405,494
2021	384	$378,192
2022	385	$403,013
2023	384	$383,741
2024	385	$424,538
2025	385	$397,739
2026 (to 29 Oct)	317	$306,432

Order volume is flat at about 385 orders a year. Revenue moves between $368K and $425K, with peaks in 2018 and 2024, but there is no sustained growth: full-year revenue in 2025 is only 2.0% above 2014. 2013 and 2026 are partial years and should not be compared directly with the full years.

14. Revenue by season

Season	Orders	Revenue	Share	Avg order value
Monsoon	1,669	$1,627,702	31.9%	$975.26
Summer	1,260	$1,319,912	25.8%	$1,047.55
Winter	1,235	$1,255,002	24.6%	$1,016.20
Autumn	836	$907,160	17.8%	$1,085.12

Monsoon is the strongest season for every product category. Autumn has the fewest orders but the highest average order value. Seasons cover different numbers of months, so volume comparisons should be read with that in mind.

15. Month and day patterns

- November is the strongest month ($459,626, 9.0% of revenue) followed by March and October; June is the weakest ($388,326, 7.6%).
- Weekdays produce 70.9% of revenue and weekends 29.1%, but this is simply 5 days versus 2: orders per day of the week are almost identical (about 714 each). Friday has the highest revenue ($759,819) and Monday the lowest ($699,721).

G. Regional Analysis

16. Revenue by region

Region	Orders	Revenue	Share	Avg order value
West	1,289	$1,345,582	26.3%	$1,043.90
North	1,276	$1,281,508	25.1%	$1,004.32
South	1,208	$1,246,641	24.4%	$1,031.99
East	1,227	$1,236,044	24.2%	$1,007.37

17. Category revenue by region ($)

Category	East	North	South	West
Electronics	411,803	472,451	487,056	458,589

Category	East	North	South	West
Clothing	402,583	364,659	348,363	416,327
Home	246,749	258,185	220,148	257,002
Beauty	174,909	186,213	191,074	213,664

Electronics is strongest in the South ($487K) and Clothing in the West ($416K). The West also leads in Beauty, while the South is weakest in Home and Clothing.

5. Excel Analysis Sheets

The workbook includes dedicated analysis sheets built with pivot tables and slicers: an Executive Summary of headline metrics, and separate Customer, Product, Time & Seasonal and Regional analysis sheets. Note that the pivot tables in the current file are filtered to the Electronics category only (1,777 orders, $1,829,899 revenue). Clearing that filter would make them match the all-category figures in this report.

6. Key Insights

- **Revenue:** total revenue is $5,109,776 across 5,000 orders, averaging $1,021.96 per order and $5,167 per customer.
- **Categories:** Electronics leads with $1,829,899 (35.8%), followed by Clothing at 30.0%. Beauty is the smallest at 15.0% but has the highest average order value ($1,059).
- **Order concentration:** the 25% of orders above $1,500 produce 54.8% of revenue; the 35% of orders under $500 produce only 8.9%.
- **Customers:** 96% of customers are repeat buyers with an average of 5 orders, and the top 10% contribute 21% of revenue.
- **Discounts:** the average discount is 18%. Deeper discounts reduce revenue per order without increasing quantity or ratings.
- **Satisfaction:** the average rating is 2.97 out of 5 and 49.5% of orders are rated negatively. Delivery speed, category, region and payment method make little difference.
- **Growth:** order volume and revenue are flat year over year, around 385 orders and $370K to $425K per full year.

- **Seasonality:** Monsoon is the top season for every category, and November is the top month.
- **Regions:** the four regions are closely balanced (24% to 26% each), with West slightly ahead.
- **Payments:** Card accounts for 46% of revenue and COD for 35%.

7. Business Recommendations

- **Fix the satisfaction problem first:** with half of orders rated negatively, investigate causes beyond delivery speed, such as product quality, accuracy of listings, packaging and returns. Add a short reason field to ratings so the drivers can be measured.
- **Review the discount policy:** discounts do not increase basket size or satisfaction. Test lower caps (for example, 20%) or targeted discounts on slow-moving products to protect margin.
- **Grow the high-value segment:** High-band orders drive 55% of revenue. Use bundles, free-shipping thresholds or upsells to move Medium orders above $1,500.
- **Turn repeat buyers into a loyalty program:** most customers already buy five times on average. Reward top spenders and re-engage the 130 customers who ordered only once or twice.
- **Break the growth plateau:** volume has been flat for over a decade. Acquisition campaigns, new categories or new regions are needed, since existing customers are already highly active.
- **Use seasonality for planning:** build inventory and marketing around Monsoon and November peaks, and run promotions in weaker periods such as June and Autumn.
- **Regional focus:** promote Electronics in the South, Clothing and Beauty in the West, and look for ways to lift Home in the South.
- **Delivery and payment:** about 37% of orders still ship in 8 to 11 days. Even though ratings do not react to speed in this data, faster delivery is worth testing as a conversion lever. Encourage Card or Wallet over COD to reduce handling and failed-delivery costs.

8. Conclusion

This project shows how a clean transactional dataset can be turned into clear business answers using structured preparation, targeted analysis and pivot-table reporting. The business has a loyal, repeat-buying customer base and a well-balanced spread across

categories and regions, but it faces three clear challenges: flat growth, discounts that do not change buying behavior, and very low customer satisfaction. Addressing these, while using the strong seasonal and high-value-order patterns, offers the clearest path to improving both revenue and customer experience.